Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 80 percent training data and the rest for testing. We trained each model for 50 epochs with a batch size of 128.

2 Results

Figure 1 shows the accuracy of CNN by dataset.

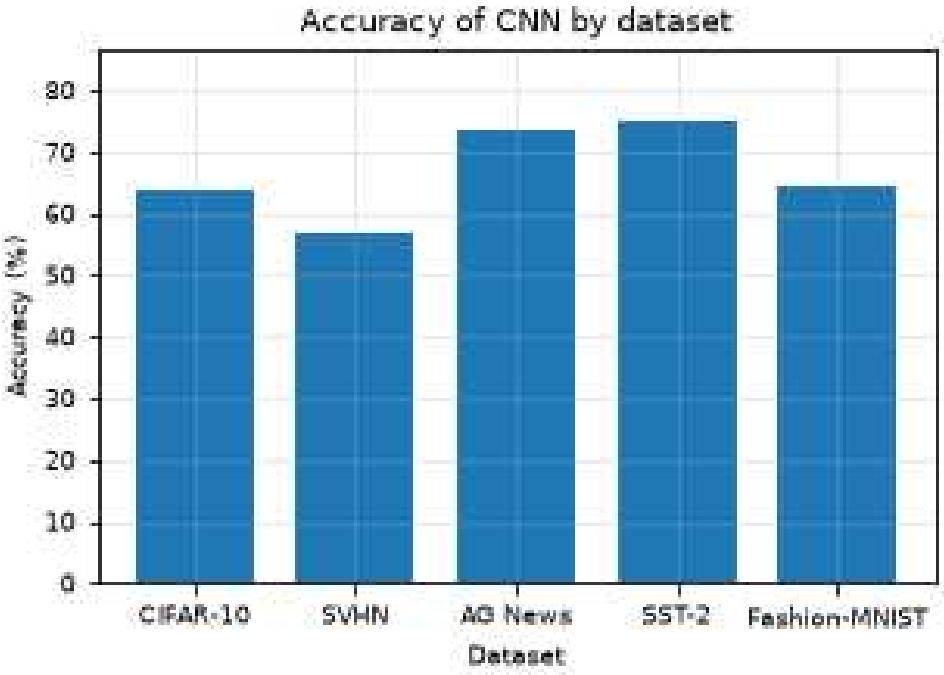


Figure 1: Accuracy of CNN by dataset for the models in this experiment.

CNN achieves 56.6% accuracy on SVHN (Figure 1). CNN scores 10.6 points higher on SST-2 than on Fashion-MNIST.

Figure 2 shows the training loss.

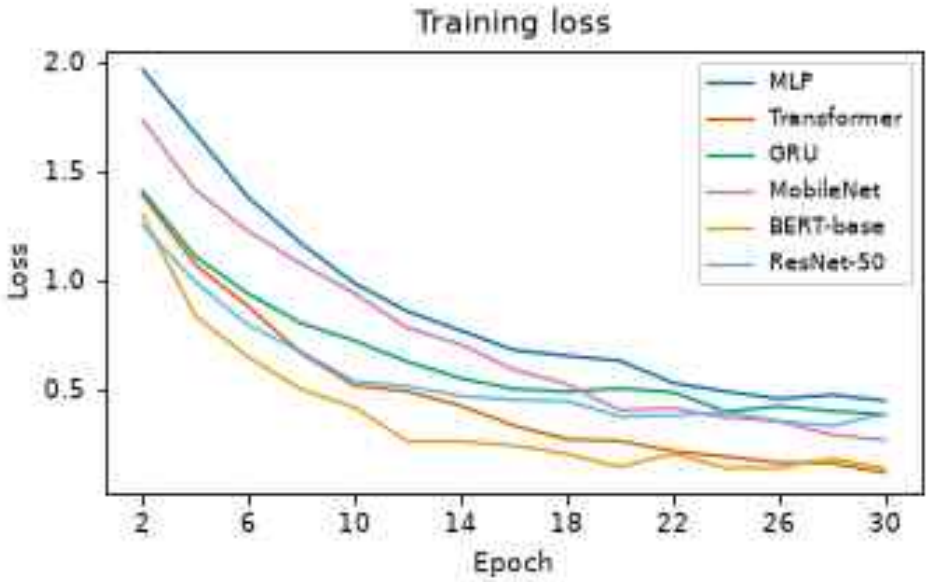


Figure 2: Training loss for the models in this experiment.

MobileNet has a loss of 1.73 at epoch 2. At epoch 26, the loss of ResNet-50 is 0.350 (Figure 2).

Figure 3 shows the accuracy during training.

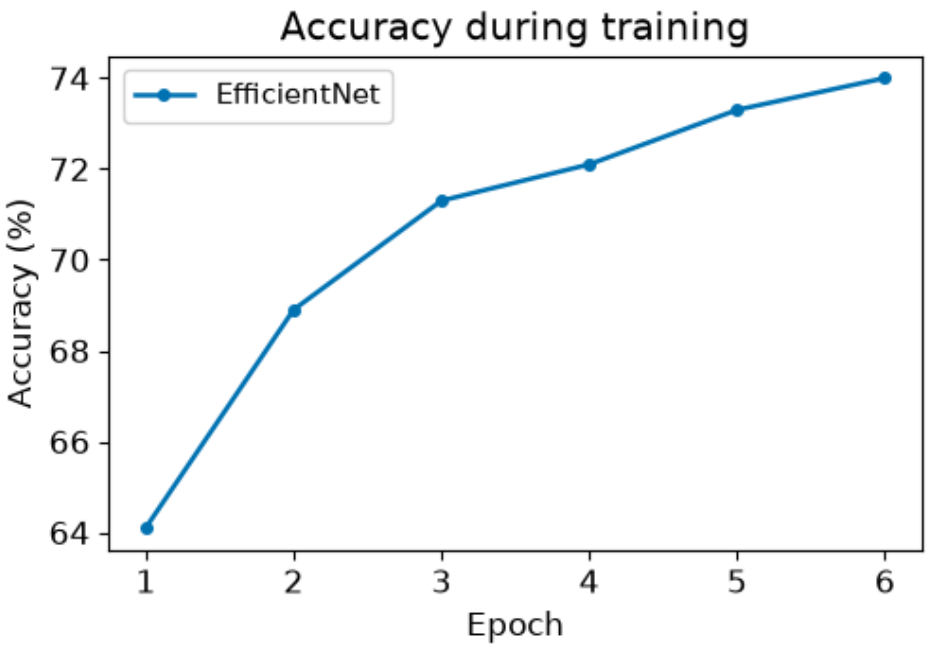


Figure 3: Accuracy during training for the models in this experiment.

From epoch 4 to epoch 6, the accuracy of EfficientNet improves by a relative 1.1%. EfficientNet scores 2.6 points higher on 2 than on 1. At epoch 4, EfficientNet achieves 72.1% accuracy (Figure 3).

3 Discussion

XGBoost reaches 76.8% accuracy on MNIST. The differences between the models are discussed in the next section.